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Assessment of Carbon Emissions Per Barrel and Decarbonization Plan in a Latin America Oilfield

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Abstract

Most operating companies have set emission reduction targets in response to regulations and stakeholder demand. Emission intensity (kgCO₂e/bbl) is the best measure for monitoring and reporting emissions of an oil and gas field. In this manuscript, we present a comprehensive and practical bottom-up approach to calculate the carbon emissions per barrel in a field. We identify all major sources of emissions, quantify their contributions to the field's overall carbon footprint, and discuss technologies to reduce the field's emission intensity.

This methodology quantifies emissions across the field development lifecycle. The analysis considers Scope 1 and 2 emissions, excluding embodied carbon products and services, and was conducted in an onshore oil producer brownfield, encompassing drilling and workover campaign, surface facilities expansion, and waterflooding operations. A detailed inventory of wellsites, wells, and equipment was considered, along with rig specifications, construction activities and energy matrix. Emissions were quantified using operational data from surface equipment, logistics, and methane monitoring technologies, and aggregated at the field level to benchmark the current carbon footprint. The emissions allocation was: energy consumption (58%), venting (6%) and flaring (36%).

Energy consumption emerged as the primary contributor to emissions, accounting for 58% of total emissions, with 71% from wells, 5% from processing, and 15% from water injection facilities. Emissions reduction can be achieved through operational efficiency (5% potential reduction), energy efficiency (5-15% potential reduction), and transitioning to cleaner energy sources. Operational efficiency involves improving field practices and adopting autonomous operations to enhance equipment reliability. Energy efficiency focuses on replacing equipment with low-energy consumption alternatives, particularly artificial lift systems and water injection pumps. Cleaner energy utilization involves integrating wellsites with the local grid to reduce diesel dependency. Venting and flaring were identified as significant contributors, with potential reductions by transporting and using associated gas for power generation. Changes in operational philosophy are necessary to transition from single-phase to multiphase flow systems. Gas monetization presents an opportunity to reduce emissions and generate revenue by producing energy for grid use. Implementing efficient artificial lift systems, improving operational practices, and utilizing associated gas

for power generation can reduce the carbon footprint by at least 20% without additional costs. The overall impact depends on scalability and expansion of these solutions within the field.

This paper proposes a novel bottom-up approach for accurately quantifying emissions and setting a carbon intensity per barrel benchmark at asset or field level using existing operational data. This method facilitates data aggregation from field to corporate level and enables the transformation of corporate decarbonization strategies into actionable plans. Integrating emissions management into routine operations empowers teams to achieve significant emissions reductions and operational improvements at minimal cost while upskilling current professionals.

Introduction

Despite the ongoing energy transition, oil and gas production is expected to remain a significant energy source for decades to come. Consequently, adapting to lower-carbon operations has become imperative for oil and gas companies, driven by increasing expectations from stakeholders, regulatory bodies, communities, and financial institutions. The International Energy Agency (IEA), in a report published in 2023, ([International Energy Agency, 2023](#)) indicates that 5.1 GtCO₂e are attributed to exploration and production activities within the oil and gas sector, equivalent to 15% of global emissions. Around 43% of total production is now subject to emissions reduction commitments considering the companies signed the Oil and Gas Decarbonization Chapter. ([Oil and Gas Decarbonization Chapter, 2024](#))

Mature fields, which account for approximately 70% of total global production, represent a good opportunity since they can achieve substantial reductions in carbon intensity, while achieving production increase and cost reductions, by decreasing energy consumption, improving production efficiency, implementing zero-venting designs, monetizing associated gas, and optimizing drilling and infrastructure operations.

However, operators still face several challenges in realizing their decarbonization strategies. These include the need to balance production targets with emissions reduction, navigating complex and evolving regulations, limited knowledge of decarbonization technologies, perceived high costs, a shortage of emissions management expertise, and difficulties in sharing emissions data across organizations.

A robust decarbonization strategy must begin with a clear and accurate assessment of current emissions per barrel of oil equivalent (BOE) produced. Yet, quantification itself presents difficulties: the industry lacks a standardized methodology for calculating emissions per barrel, with approaches ranging from life cycle assessment (LCA) models, such as OPGEE ([Adam R. Brandt, 2022](#)), to greenhouse gas (GHG) inventory methods based on frameworks like the GHG Protocol (Green House Gas Protocol). The analysis presented by ([Arthur, 2008](#)) explained the relation between LCA and GHG accounting, and particularly for oil and gas upstream operations, LCA models provide in-depth, scenario-based technical analyses but require extensive, detailed engineering and historical data, making them impractical for routine benchmarking or widespread implementation. In contrast, GHG inventory methods offer standardized, consistent, and comparable metrics aligned with investor expectations and corporate reporting requirements, though they may lack some of the granularity of LCA models to drive actionable plans.

To bridge the gap between high-level corporate strategies and actionable field-level decarbonization, this paper proposes a hybrid approach that combines inventory-based methodology that leverages operational data as inputs with the foundations of LCA models. This methodology is aligned with OGCI reporting framework ([OGCI, 2024](#)), and enables practical, asset-level tracking of emissions, ensuring both transparency and the ability to implement targeted reduction measures while supporting corporate and stakeholder reporting needs.

Proposed Methodology

This manuscript outlines a pragmatic methodology for quantifying emissions per barrel, across different stages of field development lifecycle, at the asset or field level using operational data, and for identifying key levers to create actionable decarbonization plans. The approach includes a comprehensive assessment using an LCA approach to define emissions boundaries, creation of an emissions inventory, classification of emission sources, quantification of emissions, and identification of reduction levers and solutions.

The methodology was developed and validated using real operational data from a mature field in South America, through close collaboration between the corporate sustainability team and field operations. This synergy ensured alignment between overarching corporate strategies and on-the-ground actions, enabling meaningful progress toward emissions reduction goals.

The analysis presented focuses on Scope 1 and 2 emissions, deliberately excluding embodied carbon in products and services. This approach aligns with current OGCI industry practices, as most operating companies have set reduction targets for emissions directly under their operational control, while the carbon footprint of products and services related to E&P activities is rarely tracked at the field level.

The methodology proposed here is designed to provide operators with a practical, asset-specific framework for monitoring progress against emissions targets and implementing tailored actions at the field level, considering the unique operational characteristics of each asset.

Carbon Intensity Quantification Methodology

Overview

The methodology uses the principles of OGCI guidelines ([OGCI, 2024](#)) for emissions reporting of upstream oil and gas operations, and it was applied to understand its applicability and scalability in two different mature fields located in South America. These fields are in environmental and social sensitive areas. Both fields are produced under secondary recovery and use artificial lift systems to support well production. High volumes of associated gas are present, and the area is subject to Zero flaring commitments by 2030, however the use of associated gas presents differences between the fields that result in different challenges and opportunities related to gas utilization. The main characteristics of the fields are listed in [Table 1](#).

Table 1—Main characteristics of evaluated fields

	Field 1	Field 2
Type of field	On shore	On shore
Field Lifecycle stage	Development	Development
Drilling and Development	3 WO Rigs 11 WO Jobs 27 Rigless Jobs (slickline, Coiled tubing, logging, well stimulation)	1 Drilling Rig 3 WO Rigs 8 New wells 22 WO Jobs 10 Rigless Jobs (well stimulation in producer wells)
Number of wells	39 wells: 28 Producers (deviated and horizontal) 03 Water Producers* 06 Injectors 02 Water disposal <i>*Water produced from deeper low salinity aquifer to be injected in the depleted reservoir (Piñeros, et al., 2020)</i>	82 wells: 66 Producers (deviated and horizontal) 10 Injectors 6 Water disposal
Production	Oil: 11,500 BOPD Water: 26,000 BWPD Gas: 1.9 mmcsfd Water injection: 34,000 BIPD Water disposal: 26,000 BWPD	Oil: 21,244 BOPD Water: 108,000 BWPD Gas: 9 mmcsfd Water injection: 60,000 BIPD Water disposal: 48,000 BIPD
Reservoir Depth	~10,000 ft	8,000 - 10,000 ft
Type of Oil	API 17-27	API 21-32
Well sites	04 well sites with multiple wells (well pad)	09 well sites with multiple wells (well pad) 17 well sites with one well (single well pad)
Lifting	Artificial lift (ESP)	Artificial lift (ESP)
Production mechanism	Secondary Recovery Multiple reservoirs	Secondary Recovery Multiple reservoirs
Surface facilities	01 Central processing Facility 02 Mini-Flow stations (degasification & booster) 03 Water injection system using a well-to-well closed loop configuration like the described in (Piñeros, et al., 2020, p. 16) and (Sterling, et al., 2020)	01 Central processing Facility 01 Water injection and disposal system

Methodology

The proposed methodology for quantifying the carbon intensity per barrel integrates lifecycle analysis (LCA) principles, very well established in (ISO-14040, 2006) with GHG emissions inventory accounting, specifically tailored for field-level assessment. The approach begins with a comprehensive mapping of field operations, identifying its systems, sub-systems, inflows, outflows, and their interrelations to accurately define emission sources. Operational boundaries are clearly set in line with OGCI guidelines (OGCI, 2024) and supported by Scope 1 and 2 reporting frameworks, explicitly excluding embodied carbon-upstream emissions related to manufacturing and transportation of product and services used during E&P activities—and downstream activities outside operational control, such as transportation and hydrocarbon end-use. Therefore, the methodology accounts for emissions from well production to point of sell and emissions generated from drilling and development activities conducted during the evaluated period.

The American Petroleum Institute in its latest Compendium of Greenhouse Gas Emissions Methodologies (API, 2021), proposed a classification of the major emissions categories in oil and gas operations, which was used as reference and adapted to the presented methodology to simplify the quantification process,

for the purposes of this methodology the source of emissions can be grouped into three main categories described in [Table 2](#).

A bottom-up quantification is performed by defining and assessing emission sources at the equipment and process level. Emission factors—preferably derived from field surveillance data—are applied; where such data is unavailable, (API, 2021) recommended values are used.

The calculated emissions are grouped by emissions categories (Energy consumption, gas flaring, gas venting) and aggregated to determine both total field emissions and the corresponding carbon intensity per barrel of oil produced. This enables a detailed distribution analysis of emission sources and supports the identification of targeted mitigation strategies to reduce overall carbon intensity.

This methodology was rigorously applied to two distinct oil and gas fields, utilizing operational and emission data collected over a full calendar year. The analysis encompasses all relevant activities and processes associated with field development within the defined assessment period. By employing this approach, operators can systematically close the gap between corporate-level emissions reporting and actual field performance, thereby enhancing both transparency and accuracy in emissions management.

Table 2—Emissions Source Classification

Category	Concept	Principal Sources ¹ (API, 2021) (World Bank Group, 2016)
Energy Consumption	Direct: Natural gas, fuel oil, diesel, gasoline, liquid natural gas and propane used in the Asset operation for power, heat generation, water injection, oil and gas transfer. Electricity generated on site.	Stationary Devices: Boilers, heaters, reciprocating internal combustion engines and turbines, flares, incinerators. Mobile Devices: Light and heavy vehicles, barges, trucks, construction equipment.
	Indirect: Off-site generation of electricity for on-site power	Off-site generation: Energy from fossil fuel (oil/gas power generation) Renewable sources (wind, solar, geothermal, etc)
Flaring (World Bank Group, 2016)	Routine: Flaring during regular oil production when there are no available facilities or market to economically use the associated gas.	Oil/gas separators. Oil storage tanks, tail gas treatment units, glycol dehydration facilities, produced water treatment facilities. Annular gas release.
	Safety-Flaring Flaring to ensure safe operation of the facility. Non-Routine: Flaring of gas is all flaring other than routine and safety flaring. Typically, intermittent and of short duration. It is either planned or unplanned.	Emergency shutdowns, flare's pilot flames, etc Startup following facility shutdowns, preventive maintenance and inspections, Construction activities, well maintenance activities such as acidification, wire line interventions, well testing or clean-up following drilling or well workover.
Venting	Process emissions Intentional or process-designed vented emissions released during normal operations. (API, 2021)	Storage tanks, gas blanketed tanks, gas-driven pneumatic devices, gas samplers, chemical injection pumps, drilling, well unloading, gas compressor depressurizing
	Fugitive Emissions Unintended release of gases into the atmosphere	Valves, flanges, connectors, pumps, compressor seal leaks.

Following the proposed step by step methodology and its application to the selected fields.

Assessment and Boundaries Definition. The first step of the methodology is an assessment of the processes involved in the exploration and production process of the evaluated fields in the defined time frame. The objective of this assessment is to capture the main characteristics of the field, identify the activities involved

in the upstream operations and the sources of emissions associated with each activity. The intention is to understand how emissions associated to the three main categories listed above are generated in the evaluated field(s) during the evaluated timeframe. The following steps were taken to define system boundaries, assess the processes and determine emissions sources.

E&P Flow Diagram and Boundaries

LCA concepts were used to define the specific activities and processes involved in upstream operations and how they are performed, which are grouped in systems. The inflows and outflows of each process were defined accordingly. The flow diagram in Figure 1 shows this analysis.

The boundaries are defined considering activities that are under operational control of the operating company, using the same definition of OGCI guidelines (OGCI, 2024). In this case, embodied carbon and emissions related to products and services are not accounted for as they are out of operational control. The analysis also excludes downstream activities such as transportation, refining or any other final use of the hydrocarbon produced. Note that power generation is not considered in the boundaries however, the energy consumed (that might come from this power generation system) is included in the analysis as input of the different processes.

This comprehensive boundary definition ensures that the emissions inventory is robust, transparent, and representative of actual field performance and allows a fair comparison between different fields.

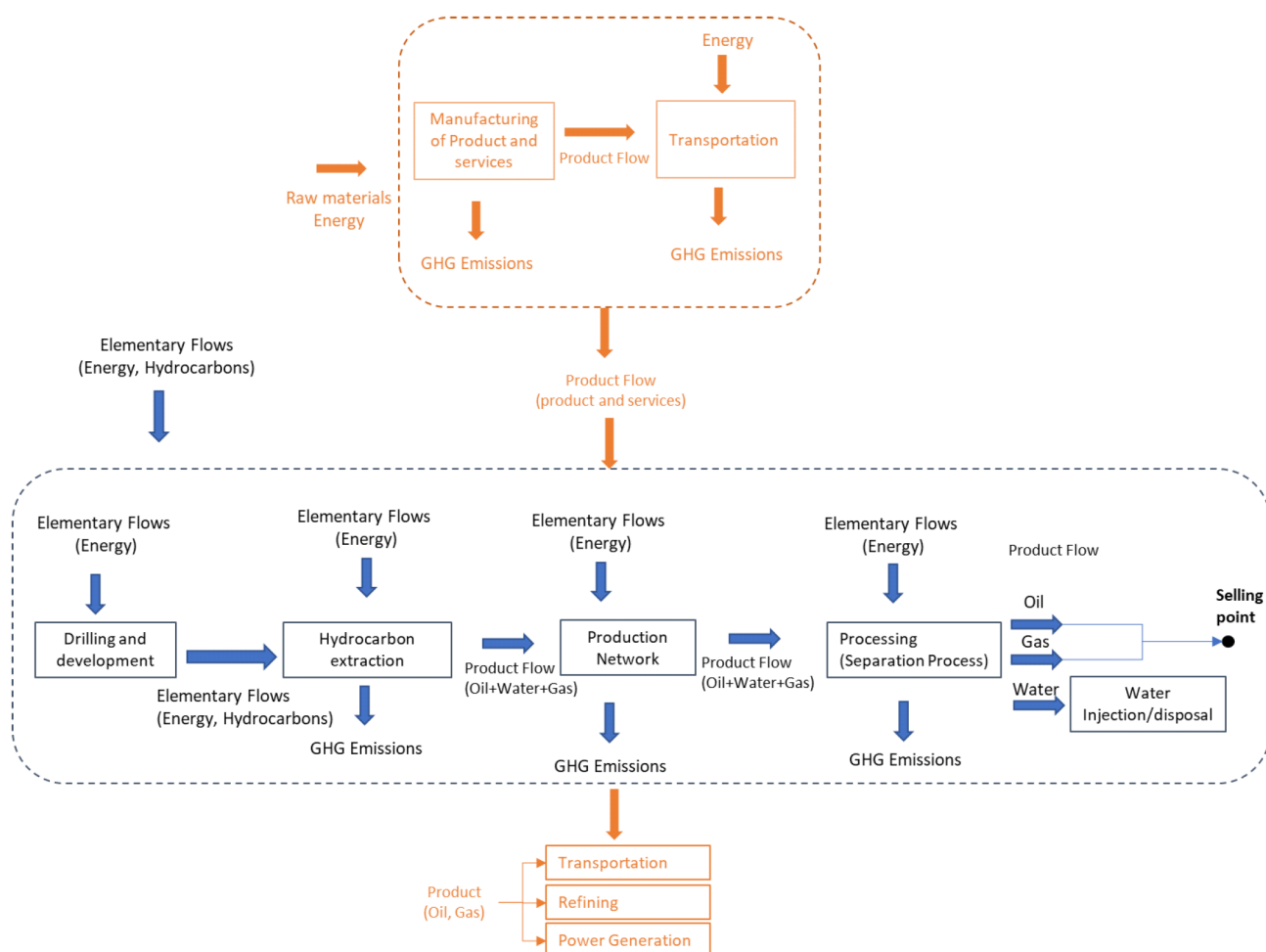


Figure 1—Exploration and Production Diagram in evaluated field with defined process boundaries

Assessment of Drilling and Development Activities

This section encompasses all operations associated with well construction, well interventions, workovers, infrastructure development, early production facilities, and transportation, each designed to enhance reservoir productivity and establish the required surface facilities to manage additional oil production. The scale and scope of these activities are inherently linked to the field's maturity: in more advanced stages, development efforts decrease, shifting focus toward production maintenance through well interventions and rigless operations.

Table 3 presents an overview of drilling and development activities in the selected fields during a calendar year. This assessment aims to delineate the full range of operations undertaken, the types of equipment deployed, job frequency, energy sources utilized, and the operational practices for flaring and venting.

Table 3—Assessment of Drilling and Development Activities

	Field 1	Field 2
Well Construction and Intervention		
Equipment		
Drilling Rigs	---	Rig 1: 2000 HP
	Rig 1: 750 HP	Rig 1: 750 HP
WO Rigs	Rig 2: 750 HP	Rig 2: 750 HP
	Rig 3: 750 HP	Rig 3: 750 HP
Fuel type	All rigs powered by diesel	All rigs powered by diesel
Fuel tracking	Daily reporting	Daily reporting
Activity		
New wells	---	8 new wells (deviated)
WO Jobs	07 Maintenance (10 days/job) 04 Recompletion (16 days/job)	10 Hydraulic Fracture (22 days/job) 08 Maintenance (7 days/job) 04 Recompletion (16 days/job)
Rigless Jobs	12 Injection Logs 05 Well stimulations 06 Slickline jobs 04 Well cleaning with coiled tubing	10 well stimulations
Infrastructure Construction	CPF commissioning Annular gas gathering systems Revamp production network	Annular gas compression to flow line New wells Wellsite preparation Revamp water management system
Transportation		
Vehicles	Vehicles for field personnel Vehicles for personnel commuting Fuel transportation Ambulance Heavy vehicles Transportation for 3rd parties	Vehicles for field personnel Vehicles for personnel commuting Fuel transportation Ambulance Heavy vehicles Transportation for 3rd parties
Fuel type	All vehicles powered by diesel	All vehicles powered by diesel
Fuel tracking	Daily logistic report	Daily logistic report

The principal activities identified include:

- Well Construction and Intervention: This category covers the drilling of new wells, workover campaigns, and rigless operations such as well stimulation, reservoir evaluation, and slickline jobs. These tasks are predominantly executed using drilling and workover rigs, as well as specialized

units for logging, pumping, and testing. The primary emissions sources are associated with fuel combustion in rig engines and ancillary equipment. According to (API, 2021), engine-driven generators on drilling rigs constitute the largest fraction of drilling-related emissions. Flaring, although limited to short intervals during flowback in workover or stimulation operations, is also a notable contributor. Additional sources include wellsite machinery—such as forklifts, cranes, and vehicles—and venting, particularly when entrained gas is released from drilling fluids during penetration of hydrocarbon zones. The magnitude of emissions is contingent upon well complexity, depth, and, crucially, the energy mix powering operations. In regions transitioning to grid-based or lower-carbon energy sources, these distinctions are essential for fair benchmarking and comparison. In the evaluated fields, emissions estimates are made at the planning stage using methodologies such as (Alexey Ruzhnikov, Aleksandr Nifantov, & Mohammed, 2023), with reconciliation against real data upon job completion.

- **Infrastructure Construction:** This encompasses wellsite preparation, expansion or upgrading of existing surface facilities, construction in new development or appraisal zones, and projects targeting water management or gas utilization. The dominant emissions source within the operational boundary is the combustion of diesel or other fuels in construction machinery (e.g., cranes, forklifts, trucks). Detailed emissions accounting relies on tracked fuel use for these activities, applying standard factors as recommended by (API, 2021). In the evaluated fields, specific equipment inventories are not easily available, however, there is a mature process in place to manage fuel records enabling a robust estimation of emissions from infrastructure development.
- **Early Production Facilities (EPFs):** EPFs enable accelerated production during the period between field discovery and the commissioning of permanent facilities. Here, the primary emissions arise from surface processing equipment—such as separators, tanks, pumps, and heater treaters—due to their energy demand, as well as process-related and fugitive gas venting. Routine flaring of associated gas is common in EPFs. (Augusto Correnti, 2022) provides a representative analysis of energy and emissions performance in such facilities. For the fields evaluated in this study, no EPFs were in operation during the assessment period.
- **Transportation** encompasses all mobile sources employed throughout routine oil and gas operations within the defined operational boundaries. This includes, but is not limited to, vehicles used for personnel movement, supply boats, trucks, and heavy-duty equipment. Typically, logistics departments oversee these activities, ensuring detailed tracking of kilometers traveled and fuel consumed, which facilitates accurate emissions estimation from transportation-related sources.

Assessment of Hydrocarbon Extraction and Processing Activities

This section provides a technical assessment of all upstream activities, from hydrocarbon extraction at the wellhead to the final processing stages prior to the point of sale, conducted across the two evaluated fields. Figure 2 and Figure 3 present schematic diagrams summarizing the principal findings and system configurations. The assessment encompasses the following stages:

- **Hydrocarbon Extraction:** The evaluation covers extraction methods from the production wells, including the deployment of artificial lift systems, chemical injection systems, management of accumulated annular gas, and the overall distribution and classification of wells. Special attention is given to mature fields where annular gas buildup is present, requiring robust gas management protocols to mitigate operational and emissions risks. (Climate & Clean Air Coalition OGMP, 2015)
- **Production Transportation in Flowlines:** The integrity and design of the gathering system are analyzed, detailing how produced fluids are directed from the well sites to the Central Processing Facilities. In Field 1, Figure 2 illustrates the use of intermediate wellpads functioning as

degasification stations, which address multiphase flow limitations in the gathering pipelines. While such configurations can alleviate backpressure and enhance throughput, they also introduce additional emission points—such as increased risk of fugitive methane emissions, localized gas venting, and flaring events—contributing to higher greenhouse gas intensities if not properly managed. Conversely, Field 2 (Figure 3) demonstrates a more streamlined approach, with direct transportation from wellsites to the main processing facility, minimizing intermediate handling and potential emission sources.

- **Processing Facilities:** The assessment includes a detailed review of the process flow diagrams and critical equipment deployed for oil, gas, and water separation and treatment. Identifying the types of equipment and their respective energy sources allows for precise mapping of stationary combustion, process venting, and fugitive emission points—enabling a granular quantification of Scope 1 and Scope 2 GHG emissions in alignment with (API, 2021) and (OGCI, 2024)
- **Produced Water Management and Reinjection:** Water handling practices are analyzed, including separation, treatment, storage, and final disposal strategies. Across both fields, produced water is primarily reinjected for subsurface disposal. In Field 1, all produced water is injected into non-productive reservoirs, while Field 2 employs a waterflooding strategy, reinjecting approximately 55% of produced water into depleted reservoirs to enhance secondary recovery, with the residual disposed in non-productive zones.
- **Associated Gas Utilization and Flaring Reduction:** The handling and disposition of associated gas streams are key focus areas. In Field 1, most of the associated gas is directed to flare at the processing facility, representing a significant emissions source and an opportunity for mitigation. Field 2 takes a more progressive approach—compressing and exporting a portion of the associated gas for sale, while still flaring the remainder. This split highlights actionable opportunities for emissions reduction and value creation through expanded gas capture and monetization, aligning with industry zero routine flaring objectives.

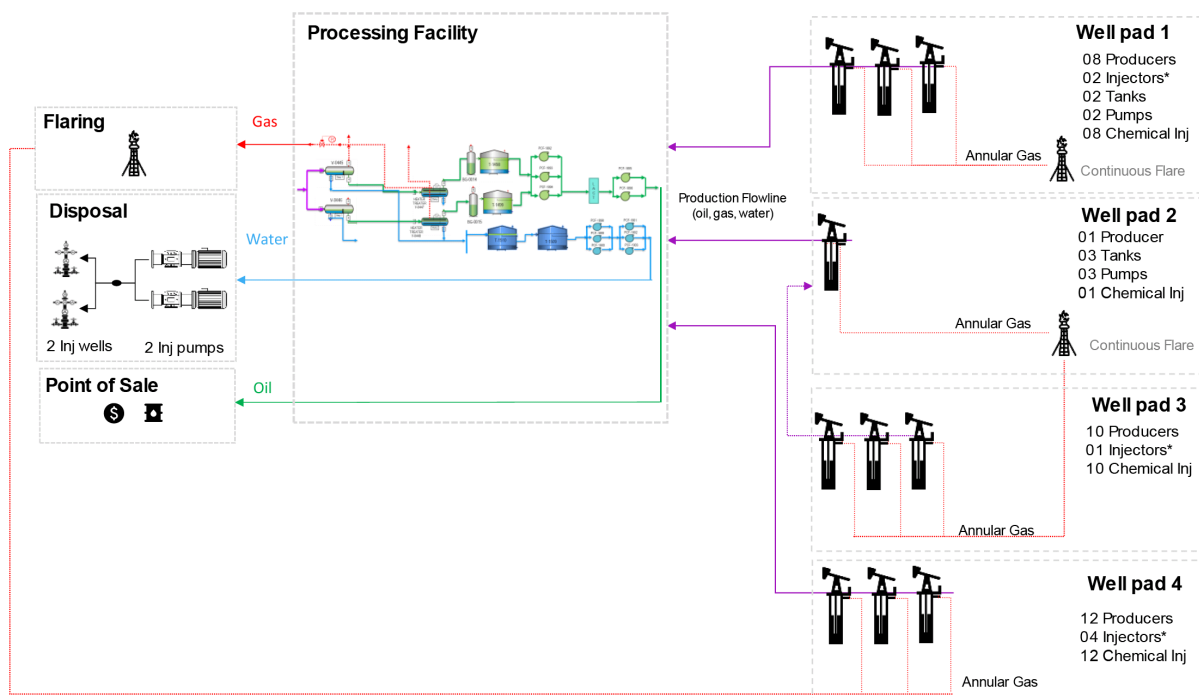


Figure 2—Assessment of Hydrocarbon Extraction and Processing – Field 1

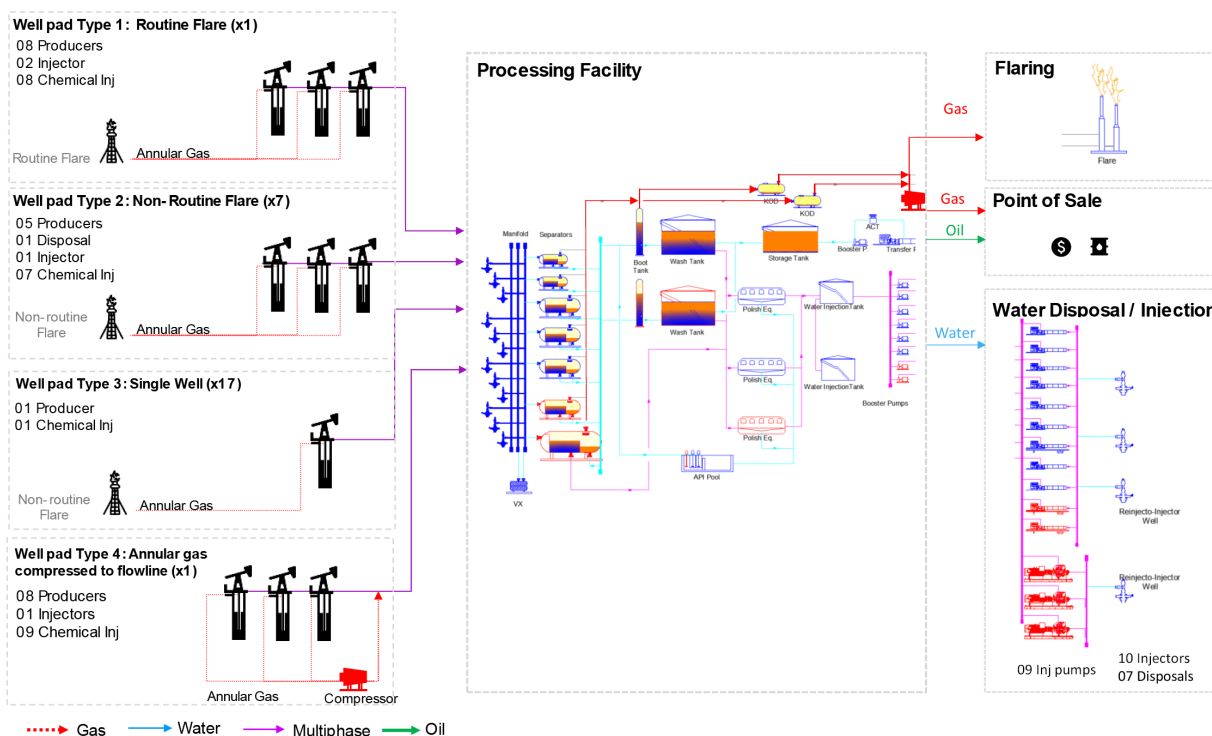


Figure 3—Assessment of Hydrocarbon Extraction and Processing – Field 2

By systematically evaluating the technical configuration and operational practices across these domains, we can delineate the principal sources and magnitudes of emissions—energy consumption, flaring, venting, and fugitive releases—within the defined system boundaries.

Assessment of Energy network and Power systems

In certain oil and gas fields, operators are responsible for in-house electricity generation to meet the demands of critical production equipment, such as artificial lift units, injection pumps, and transfer pumps. Conducting a thorough evaluation of the power generation system is essential for delineating the emission boundaries associated with this energy source, as well as for understanding the energy mix, associated emission factors, and opportunities for decarbonization through the adoption of lower-carbon or renewable alternatives.

Figure 4 illustrates the electrical network architecture pertinent to the evaluated fields. Isolated remote fields typically rely on autonomous power generation infrastructure, utilizing a combination of on-site oil, associated gas, and diesel-fired generators to maintain operational continuity. In contrast, the evaluated fields are embedded within a localized power grid, wherein centralized power generation hubs supply electricity via an integrated network. These hubs employ a diversified fuel portfolio—including oil, diesel, and associated gas—to optimize energy availability and reliability. Gas is aggregated from upstream processing facilities and directed to select power hubs, while others predominantly use oil and diesel, depending on resource accessibility. Notably, a segment of the supplied energy is imported from the national grid, introducing additional considerations regarding grid emissions factors, energy independence, and the potential integration of cleaner energy sources.

This system-level analysis reveals that, although power generation is distributed across the evaluated fields, the direct emissions associated with these generation activities fall outside the defined operational boundaries of the current emissions assessment. This is due to the shared nature of the power network, where electricity produced at generation hubs is utilized by multiple fields. As a result, only the emissions attributable to on-site energy consumption are allocated within the system boundaries, in accordance with the principles of life cycle assessment (LCA) and operational control guidelines (API, 2021). These energy

consumption-related emissions are quantified as part of the elementary flows within each system delineated in the E&P process flow diagram presented in Figure 1.

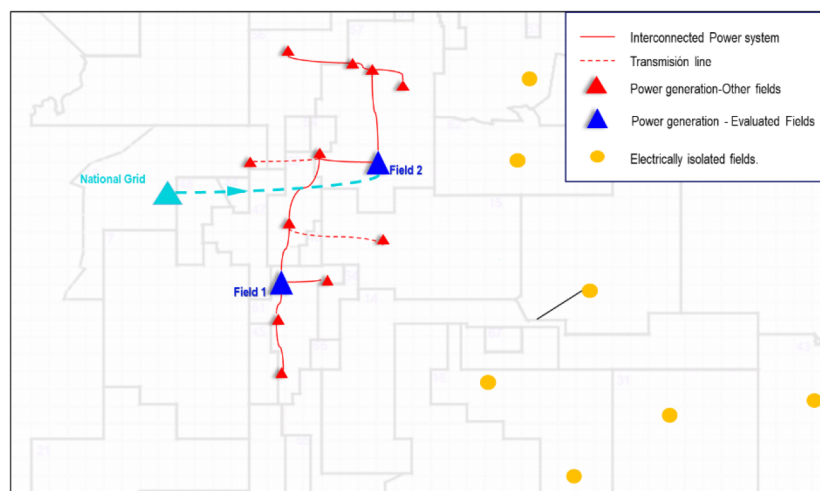


Figure 4—Electrical network in evaluated fields

Assessment of Gas Flaring

Recent findings from (World Bank Group, 2025) reveal that global gas flaring increased for a second consecutive year in 2024. Despite international efforts, the routine flaring of associated gas remains widespread, largely due to technical and economic constraints. Factors such as limited field infrastructure, low gas volumes at dispersed well and processing sites, variable gas composition, and challenging operational conditions delay the monetization of associated gas streams. As a result, significant quantities continue to be flared due to their relatively low market value and the high cost of infrastructure investment.

In the evaluated fields, associated gas production averages approximately 5 mmscfd, Table 4 details the distribution of flared gas across the operational sites. The dispersion of flaring at multiple locations within the field poses a significant logistical and economic challenge. Aggregating all associated gas to a single collection point would involve substantial infrastructure upgrades. Nevertheless, targeted solutions—such as localized gas compression into flowlines or the strategic development of nearby gas gathering points—could facilitate the reduction or elimination of smaller, decentralized flares and enhance the capture of gas for productive use.

Table 4—Gas Flare distribution (Volume of gas flared in mmscfd)

Field 1		Field 2	
Central Processing Facility	1.16	Central Processing Facility	1.86
Well pad 1	0.52	Well pad Type 1	1.10
Well pad 3	0.17	Well pad Type 2 & 3 (24 wellsites)	0.04
Heater treaters in CPF	0.05	Hydraulic Fracturing (10 jobs)	0.004
Total	1.90	Total	3.00

Further technical consideration arises from reservoir depletion in certain wells, which leads to increased gas liberation within the wellbore because of pressure drawdown. This liberated gas accumulates in the annulus, elevating casing pressures and adversely affecting the performance of downhole pumps and well productivity—a phenomenon thoroughly described in the (Climate & Clean Air Coalition OGMP, 2015).

Where annular gas infrastructure is lacking, operators typically vent this gas directly to the atmosphere by bleeding annular pressure, a practice that contributes to elevated venting emissions. However, in the fields under assessment, current operations have adopted annular gas gathering systems that channel this gas either to a designated flare at the well pad or, where feasible, compress it into the flowline for co-processing with the multiphase stream at central facilities, as illustrated in [Figure 2](#) and [Figure 3](#).

Flaring associated with well construction and intervention activities was observed exclusively during hydraulic fracturing operations in Field 2, specifically during the flowback phase ([Abdul Muqtadir Khan, 2024](#)). Although flare volumes in these operations are not measured, they can be estimated based on well production rates during flowback and the gas-oil-ratio (GOR). When compared on an annual basis to flaring volumes associated with regular production operations, these contributions are negligible for field emissions accounting purposes.

In Field 1, a small volume of associated gas is used by the heater treaters as part of the dehydration process shown in [Figure 2](#), the combustion of this volume of gas is included in this section ([Table 4](#)).

Assessment of Venting Sources

According to the latest ([International Energy Agency, 2025](#)) report, annual methane emissions from upstream oil and gas activities are estimated at approximately 60 million tonnes, underscoring the critical need for rigorous venting analyses to quantify current emissions and support pathways toward net-zero methane targets.

In this evaluation, industry-standard frameworks—such as those set by the ([OGCI, 2024](#))—and venting estimation methodologies recommended by ([API, 2021](#), p. section 6) were adopted to ensure methodological robustness. [Table 5](#) provides a consolidated overview of identified venting sources across the assessed fields.

Table 5—Venting Sources in Evaluated Fields

	Field 1	Field 2
Drilling and Development	Workover Jobs	Drilling activities Well completion Workover Jobs
Hydrocarbon Extraction	Fugitive emissions (wellheads, annular gas gathering systems, valves)	Fugitive emissions (wellheads, annular gas gathering systems, valves)
Production network	Oil Storage Tank Fugitive emissions flow lines	Fugitive emissions flow lines
Oil and Gas processing	Oil Storage Tank Separators Fugitive emissions	Oil Storage Tank Compressor venting Separators Fugitive emissions
Water injection	Produced Water Tank	Produced Water Tank

It is essential to recognize that venting source identification is more complex than quantifying flaring or energy consumption, as methane emissions from venting are typically hidden and not visible. For flaring and energy consumption, operational data facilitate direct calculations; however, accurate venting quantification requires advanced technical expertise, particularly regarding surface production facility design and operation. This includes familiarity with equipment specifications and proficiency in applying various estimation techniques tailored to operational realities.

In the evaluated areas, methane monitoring campaigns utilizing drone technology were conducted. While drone-based measurements offer valuable data for venting quantification and serve as important reference

points, it is imperative to contextualize these readings: drone surveys provide static, time-specific snapshots that may not fully represent dynamic operational conditions or temporal variability in venting rates.

Emissions inventory. Once the assessment is completed, the activities and processes identified within the defined boundaries are systematically grouped into systems. In this methodology, a system is understood as an interconnected set of elements working together to perform a function or produce a product. Each system contains sub-systems, creating a hierarchy that enables efficient data aggregation at the field level. This structure allows for detailed tracking of each sub-system independently, so that as the quantification process matures and more information becomes available, additional detail can be incorporated without disrupting the overall calculations.

For example, within System 2, the artificial lift sub-system includes various downhole pumps installed in individual wells. In the case of Field 1, we possess detailed, well-by-well data—such as the well type, the specific electric submersible pump (ESP) used, and the energy consumption for each. In contrast, Field 2 only provides the total energy consumed by the artificial lift system, without granular well-level information. As more data becomes accessible in the future, these details can be seamlessly added without affecting emissions calculations for other sub-systems.

Figure 5 illustrates the proposed approach for grouping activities and processes identified during assessment. In this section, the inventory lists the number of equipment and activities, along with the emissions they generate. Once all activities, processes, and equipment are organized by category, the resulting field inventory appears in Table 6, detailing each sub-system's components and their associated emissions. It is important to note that a single sub-system and its elements may contribute to multiple emission categories. For instance, the drilling activity sub-system produces emissions related both to energy consumption—from diesel used by the rig—and to venting and flaring from the drilling process itself.

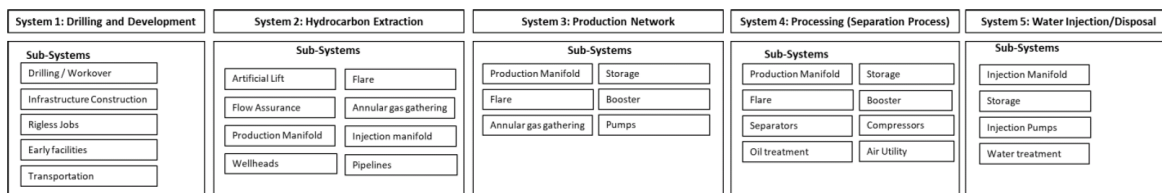


Figure 5—Proposed Systems to group activities and create field emissions inventory

Table 6—Emissions inventory of evaluated fields

System	Sub-system	Emissions generated			Inventory	
		Energy Consumption	Flaring	Venting	Field 1	Field 2
Drilling & Development	Drilling Rig	X		X	3 Rigs (11 Jobs, No Hydraulic Fracture) 3 projects 27 Jobs Light and heavy vehicles	1 Rig (8 New wells)
	WO Rig	X	X	X		3 Rig (22 Jobs, 10 Hydraulic fracture)
	Infrastructure Construction	X				5 Projects
	Rigless Jobs	X				10 Jobs
	Transportation	X				Light and heavy vehicles
Hydrocarbon Extraction	Artificial Lift	X			28 ESP	66 ESP
	Flow Assurance (Chemical injection)	X			43	102
	Production Manifold			X	5	26
	Wellheads (Casing gas vents)			X	28 Oil Producers	66 Oil producers
	Flare		X		3	2
	Annular gas Gathering			X	2	26
	Injection manifold				8	9
Production Network	Production Manifold			X		NA
	Flare		X			NA
	Annular gas Gathering			X	2	NA
	Storage			X	5	NA
	Boosters & Pumps	X			5	NA
Production Processing	Separators			X	2	4
	Oil treatment	X			13	22
	Storage			X	3	2
	Pumps & Boosters	X			6	5
	Compressors	X		X		1
	Flare		X		1	1
	Production Manifold			X	1	1
Water Injection/Disposal	Injection manifold				3	9
	Storage			X	1	2
	Injection Pumps	X			2	9
	Water treatment	X			10	25

With the emissions inventory prepared and categorized, the process of aggregated quantification becomes straightforward and transparent.

Carbon Intensity Quantification.

Energy Consumption

As was explained in the assessment session two main energy sources were identified:

1. Electricity sourced from the local power grid primarily powers equipment belonging to Systems 2, 3, 4, and 5—namely hydrocarbon extraction, production networks, processing, and water injection/disposal operations. Emissions associated with electricity use are calculated by multiplying total consumption by the relevant power grid emission factor, as detailed in [Table 7](#), which shows the calculation of the emission factor according to the fuel types used in electricity generation.
2. Diesel is consumed by mobile combustion units, including drilling and workover rigs, rigless units, construction equipment, and all vehicles enumerated in [Table 3](#). To quantify emissions from diesel use, the total volume burned is multiplied by the corresponding emission factor. For these calculations, an emission factor of 0.002713 tCO₂e per liter of diesel is applied, ensuring uniformity and transparency throughout the inventory process. This emission factor is aligned with calculation methods proposed by (API, 2021) converted to tCO₂e per liter.

Table 7—Calculation of emission factor of the field power grid

Type of Energy	Power by type of energy (MW) ²	Distribution (%)	Emission Factor (tCO ₂ e/MWh)	Emissions factor by type of fuel - Source
Crude	132	35%	0.700	(Nicholson, 2021)
Diesel	166	44%	0.877	(Nicholson, 2021)
Gas	34	9%	0.622	(Nicholson, 2021)
National grid import	45	12%	0.120	Energy and Mining Ministry, Ecuador, 2023 (Environmental Ministry)
Total	378	100%	0.738	

Emissions associated with gas flare

The quantification of emissions from gas flaring adheres to (API, 2021) guidelines. In the assessed fields, direct measurement of flare gas volume is not available; therefore, volumes are estimated based on oil production rates and the field gas-oil ratio (GOR), as reported by the operator—a common methodology in mature fields with associated gas (API, 2021). Gas composition data is obtained via recent gas chromatography analyses, providing critical input for evaluating potential gas utilization strategies. A flare combustion efficiency of 98% is assumed, supported by prior measurements using LiDAR technology. [Table 8](#) details the parameters applied in the flare emissions calculation, referenced in [Equation 1](#).

Table 8—Parameters used to calculate flare emissions

	Field 1	Field 2	Source
CO ₂	5.19%	15.49%	Gas Chromatography
CH ₄	67.27%	47.37%	Gas Chromatography
Combustion efficiency	98%	98%	LiDAR
Calculated tCO₂e per mmscf of gas flared	78.83	61.66	Equation 1

The evaluation indicated that gas flared in the assessed fields has a consistent composition across all flaring points. Consequently, Equation 1 was applied to estimate emissions per million standard cubic feet (mmscf) of gas flared, with the results summarized in Table 8. These emission factors were then multiplied by the gas volume at each flaring site, as listed in Table 4, to calculate total gas flaring emissions.

Equation 1: Calculation of Gas Flare emissions:

$$GHG \text{ combustion in flare} = CE \times V \text{ flared gas} \times CH_4 \text{ content} \times NG \text{ combustion factor} \quad (81.21) \quad (1)$$

$$GHG \text{ from } CH_4 \text{ unburned} = (1 - CE) \times V \text{ flared gas} \times CH_4 \text{ content} \times 84 \times 20 \quad (2)$$

$$Total \text{ GHG Flare emissions} = (1) + (2)$$

Where:

V : Total Volume of gas flared during the evaluated period (mmscf)

CE : Combustion efficiency (%)

GHG: Greenhouse gases emissions in tCO₂e

NG: Natural gas

Emissions associated with gas venting

The quantification of emissions from gas venting is conducted in accordance with (API, 2021) guidelines. Unlike flaring, venting is rarely monitored continuously and is difficult to track due to its invisible nature. As a result, emission calculations primarily rely on indirect estimates based on factors and using operational procedures and equipment specifications. For the methodology presented in this manuscript, the estimation methods and factors proposed by (API, 2021, p. section 6) were used.

Vented gas—whether released intentionally through process design (process emissions) or unintentionally via leaks (fugitive emissions)—retains the same composition as the reservoir gas entering the production stream. Therefore, its properties are considered consistent across the evaluated system. Like flared gas emissions, Equation 2 is used to calculate emissions per million standard cubic feet (mmscf) of vented gas. Since the volume of vented gas is not continuously monitored across the systems described in Figure 5, estimates are based on available direct measurements and missing data is supplemented using the methodology detailed in (API, 2021, pp. 242-411, section 6), once the vented gas volume is determined for each source, it is multiplied by the emission factor from Equation 2 to quantify total emissions.

Table 9—Parameters used to calculate venting emissions

	Field 1	Field 2	Source
CH ₄ concentration (%)	67.27	47.37	Gas Chromatography
CO ₂ concentration (%)	5.19	15.49	Gas Chromatography
Methane GWP x CO ₂ by weight	84	84	(IPCC, 2023)
CH ₄ conversion to Tonnes (Tonnes of methane per mmscf methane)	20	20	(IPCC, 2023)
CO ₂ conversion factor to Tonnes (Tonnes of CO ₂ per mmscf CO ₂)	51.94	51.94	(IPCC, 2023)
Calculated tCO₂e per mmscf of gas vented	1132.83	803.86	Equation 2

Equation 2: Calculation of gas vented emissions:

$$GHG \text{ from } CH_4 \text{ venting} = V \text{ vented gas} \times CH_4 \text{ content (\%)} \times CH_4 \text{ GWP (84)} \times 20$$

$$GHG \text{ from } CO_2 \text{ venting} = (V \text{ vented gas} \times CO_2 \text{ content (\%)} \times 51.94) + (V \text{ vented gas} \times CO_2 \text{ content (\%)} \times 51.94)$$

Where:

V : Total Volume of gas vented during the evaluated period (mmsf)

GHG: Greenhouse gases emissions in tCO₂e

GWP: Global warming potential

During drilling operations, venting typically occurs when gas is released from the drilling fluid upon entering the production zone. This practice is often necessary for safety and operational reasons, such as managing pressure surges during drilling, and it involves the controlled release of natural gas directly into the atmosphere. Measuring the exact volume of released gas is challenging.

While venting is sometimes essential in certain drilling operations, it is crucial to minimize its frequency and environmental impact through careful planning, adherence to operational best practices, and regulatory oversight. In the Field 2, eight wells were drilled, and the estimation method used was based on simplified emission factors recommended by (API, 2021, pp. 6-3). Specifically, the factor for onshore wells using water-based mud (2,857 scf gas per drilling day) was applied. With 25 drilling days per well, venting emissions associated with drilling activities are 459 tCO₂e. The result is shown in Table 10.

Table 10—Vented gas Emissions Calculation in tCO₂e/yr

	Factors	Field 1	Field 2	Source
Well Construction Activities				
Drilling	2857 scf gas/drilling day	---	459.3	(API, 2021, pp. 6-3)
Workover without Hydraulic Fracturing	122 scf gas/ workover	1.52	1.17	(API, 2021, pp. 6-22)
Workover with Hydraulic Fracturing	39,357 scf / workover	---	319	(API, 2021, pp. 6-13)
Hydrocarbon Extraction				
Wellhead (leaks)		2,030	1,214	From Drone survey
Annular gas gathering (leaks)		250	110	From Drone survey
Production network				
Storage	0.00525 KgCH ₄ /bbl crude	547	---	(API, 2021, pp. 6-69)
Fugitive emissions		589	450	From Drone survey
Production processing				
Storage	0.00525 KgCH ₄ /bbl crude	2,012	3,420	(API, 2021, pp. 6-69)
Separators		200	654	From Drone survey
Compressor		--	1,295	Drone survey & Blowout Estimations
Fugitive emissions		590	625	From Drone survey
Water injection/disposal				
Water tank	0.0015 tonne CH ₄ /1000 bbl produced water (P. Separator 50 psi)	1196	4,967	(API, 2021, pp. 6-76)

For workover operations, venting can occur during the removal of production tubing, causing gas to escape into the atmosphere. In the absence of direct data, recommended emission factors were used to estimate venting during workovers as outlined in (API, 2021, pp. 6-13 and 6-22)

During the assessment, it was determined that the main sources of venting emissions in hydrocarbon extraction, production networks, processing facilities, and water injection or disposal are fugitive emissions across the system (wellheads, flowlines, valves, etc), and process venting emissions associated with storage tanks, separators and compressors. Practices that are common in some mature fields such as associated gas and casing gas vents were not identified in the evaluated area, as it was explained before, gas built up in annular space is gathered and mixed with the associated gas from the production stream to be separated and flared in the production processing facilities.

Estimates for venting in hydrocarbon extraction, production networks, processing facilities, and water injection or disposal are derived from drone surveys conducted in the evaluated field. These surveys served to validate baseline methane emissions and provide a clearer understanding of operational realities. Methane sources identified by drones were cross-referenced with previous estimations, showing strong correlation for key sources such as tanks and separators. Where direct data was available, survey results were extrapolated to similar wellsites. While drone surveys offer only a snapshot of field conditions, they are valuable for validating calculations and identifying new emission sources previously undetected.

Non-routine venting emissions, which may arise from maintenance, equipment failures, and other irregular activities, were estimated as fugitive emissions. These calculations followed (API, 2021) guidelines and incorporated drone data for additional validation and adjustment.

Carbon Intensity: Baseline Results

Following the established methodology, the carbon intensity for the selected fields was quantified and is presented in Table 11. Figure 6 and Figure 7 further illustrate the distribution and contribution of each emission category, revealing the main sources of greenhouse gas emissions. Notably, the distribution varies between the two fields, particularly concerning flaring emissions. Field 2 demonstrates a reduction in flaring emissions due to its strategy of utilizing a portion of the associated gas—compressing it at the processing facility and transferring it to the point of sale for further use. In contrast, Field 1 flares all associated gas, though the accumulation at the processing facility presents a potential opportunity for enhanced gas utilization.



Figure 6—Emissions results and their distribution for Field 1

Table 11—Total Emissions and Carbon intensity of the selected fields

System	Sub-system	Field 1 Emissions tCO ₂ e/yr			Field 2 Emissions tCO ₂ e/yr		
		Energy Consumption	Flaring	Venting	Energy Consumption	Flaring	Venting
Drilling & Development	Drilling Activity	-			5,674		459
	Completion & WO Activity	2,676		1.52	1,309	900	319
	Infrastructure Construction	1,511			883		
	Rigless Jobs	3,026			682		
	Transportation	941			539		
Hydrocarbon Extraction	Artificial Lift	46,073			96,268		
	Flow Assurance (Chemical injection)	691			13		
	Production Manifold						
	Wellheads (Casing gas leaks)			2,030			1,214
	Flare		4,892			25,657	
	Annular gas Gathering			250	776		110
Production Network	Injection manifold						
	Production Manifold						
	Flare		14,963				
	Annular gas Gathering						
	Storage			547			
	Fugitive emissions			589			450
Production Processing	Boosters & Pumps	2,080					
	Separators			200			654
	Oil treatment	68					
	Storage			2,012	3		3,420
	Pumps & Boosters	4,526			2,586		
	Compressors						1,295
	Heater Treaters		1,381				
	Flare		33,377			41,861	
Water Injection/Disposal	Fugitive emissions			590			625
	Production Manifold						
	Injection manifold						
	Storage			1,196			4,470
	Injection Pumps	423			29,094		
	Water treatment	6			3		
	Total by Emission Category (tCO₂e)	62,021	54,612	7,415	137,831	68,418	13,016
	%	50%	44%	6%	63%	31%	6%
	Field Total emissions (tCO₂e)			124,048			219,265
	Total Production (BOE)			4,562,500			7,754,060
	Carbon Intensity (KgCO₂e/BOE)			27.19			28.28



Figure 7—Emissions results and their distribution for Field 2

Water management emerges as a significant factor, with Field 2 handling substantially higher liquid volumes due to its elevated water cut (83%), which leads to increased energy consumption emissions. The data shows that the dominant contributor to energy consumption emissions in both fields is the artificial lift

system, followed by the water injection system in Field 2, reflecting the large volumes of water requiring disposal. The implementation of a well-to-well closed loop system in Field 1 (Dmitry Pleshkov, 2022) has a positive impact, notably reducing emissions associated with energy consumption during waterflooding operations.

Venting emissions are more prominent in Field 1, primarily due to the configuration of its production network, which utilizes degasification systems at multiple wellsites.

These insights serve as a foundation for targeted emissions reduction strategies and further optimization of carbon intensity across the fields.

Approach to reducing Carbon intensity

Effectively managing carbon intensity is not merely a compliance exercise; it represents a strategic lever to drive operational excellence and ensure the long-term resilience of assets. A comprehensive approach that integrates emissions management with resource optimization delivers significant advantages, both in terms of sustainability and value creation.

A cost-efficient transition to lower-carbon operations should be embedded throughout the asset lifecycle. Rather than perceiving emissions reduction as a limitation, operators are encouraged to proactively pursue performance improvement initiatives that address process inefficiencies, reduce operating costs, and accelerate production. The adoption of digital technologies and modern operational practices enables reductions in well failures, non-productive time, and cycle durations, providing measurable improvements in emissions performance and overall profitability.

Production enhancement plans should focus on maximizing output with a lower carbon footprint, incorporating low-carbon solutions from the design phase onward to limit emissions at the source. Initiatives aimed at reducing energy consumption contribute directly to lower emissions and offer substantial cost savings by targeting key sources of greenhouse gases.

Innovation is also essential for unlocking additional value within emissions management. By leveraging emerging technologies and adopting new commercial models, operators could monetize associated gas, diversify revenue streams, and establish strategic partnerships. In this way, emissions management serves not only as an environmental imperative but as an enabler of operational optimization and long-term competitiveness.

Potential Solutions to reduce carbon intensity

Reducing carbon intensity in upstream oil and gas operations requires a multifaceted approach targeting the main emission sources: energy consumption, flaring, and venting. This section outlines relevant examples of key technologies and operational strategies implemented in similar fields that have demonstrated effectiveness in lowering emissions across these categories.

Technologies associated with Energy Consumption

Highly efficient downhole pumps. Artificial lift systems are the largest contributors to energy-related emissions in mature fields. The implementation of new generation of motors such as PPM, compact wide range and encapsulated motors in similar fields has shown between 8 to 30% reduction in energy consumption per barrel produced, while increasing oil recovery and reduce well interventions. (M. N. Saputra & B. A. Rahmanto and H. Anggaraini, 2025)

Smart Horizontal Pump Systems. The Horizontal Pump System (HPS) stands as the second largest contributor to energy-related emissions in mature oil fields, following artificial lift systems. HPS play an essential role in transporting and injecting formation water from processing stations to disposal and waterflooding wells under high pressure. This vital function not only maintains fluid management but also sustains oil production, especially in brownfield operations.

Yet, operational challenges such as scale buildup and solids accumulation frequently disrupt HPS performance, leading to increased energy use, system failures, and production losses. Addressing these inefficiencies, the deployment of digital solutions—leveraging sensors, edge/cloud computing, and advanced analytics—has markedly enhanced the reliability and efficiency of pump operations. Real-time monitoring allows for rapid detection and resolution of anomalies, minimizing downtime and unnecessary field interventions.

The impact of these digital strategies is significant: system failures were reduced by 70%, production losses decreased by 93%. By optimizing pump reliability and reducing power consumption by 10%. (Santiago Davalos, 2024)

Smart Production Operations. The deployment of IoT, edge/cloud computing, and AI/ML to automate production operations such as testing, chemical injection, and artificial lift optimization has been proved in similar fields enabling more sustainable, and improving operational efficiency by 60%, reduced field trips by 57%, and significantly lowered emissions from diesel use and maintenance activities (Paola Rueda, 2023). Having remote and autonomous operations has also significantly impacted the reduction in well failures, avoiding the necessity of rigless and rig interventions, reducing the related GHG emissions by less diesel consumption, materials utilization and wastes generation

Electrical Balance. In the evaluated fields an innovative digital solution for energy balance was developed that integrates IIoT, real-time data acquisition, and predictive modeling. The main strength of this system lies in the online quantification of carbon emissions, considering the energy matrix and fuel sources, enabling continuous tracking of the carbon footprint. This solution also monitors electrical parameters in wells and power systems, calculates efficiency indicators (kW/BFPD), and classifies wells by energy consumption and production to pinpoint inefficiencies. Additionally, it includes simulation tools for future load scenarios and reliability schemes, delivering energy savings, emission reductions, and improved sustainability in alignment with the Sustainable Development Goals. (Luis Tilleria, 2025)

Rig Power management. Rig fuel consumption is the main source of emissions in well construction activities. The adoption of Load Dependent Software System (LDSS) and Intelligent Power Management in drilling operations addresses fuel efficiency and emissions reduction by optimizing engine usage on rigs. LDSS leverages real-time load sensing and digital controls to regulate engine activity, ensuring engines run only when necessary. This technological solution led to a 10% reduction in diesel consumption per well, a 20% decrease in engine operating hours, and approximately 200 tonnes less CO₂ emissions per rig each year. The approach not only improves drilling efficiency and reliability but also delivers significant cost savings in fuel and engine maintenance, while opening opportunities for further power plant improvements—directly supporting the sustainability goals of upstream operations. (Chandharabhanich, et al., 2024)

Autonomous Drilling. A critical component to reduce the emissions of drilling operations is by improving performance and optimizing the process to reduce energy consumption. A digital solution was implemented to improve efficiency, reduce Non-Productive Time (NPT) and to reduce emissions. The drilling rig was adapted for rig autonomous operations enabled by digital sensors, IoT, cloud computing and AI/ML. The rig automation system uses a goal-based approach to handle constantly changing drilling conditions like drilling dysfunctions, formation changes and repetitive tasks such as downlinking, and pre- and post-connection procedures, without having constantly rely on human intervention. This implementation has allowed the drilling team to operate the rig in 85% of autonomous mode during the operations activity for each well, improving efficiency and resulting in the reduction of 1.2 days of operations, representing less diesel and energy consumption, minimizing human errors and uncertainty of dynamically changing operations, and impacting significantly in the reduction of carbon footprint, operational risk and cost, aligned with the sustainability objectives and goals of the management team for field development (Etcheverry, et al., 2023)

Flaring Reduction

Annular Gas gathering systems. Decentralized flaring has been effectively addressed through the installation of annular gas compressors and gathering systems. This approach, now deployed in the evaluated fields, has enabled the decommissioning of flares, improved production rates, and prevented substantial gas emissions. In mature, depleted fields, increased gas production due to lower bottomhole pressures often impacts pump performance. Proactive integration of these systems into field development plans not only reduces operational disruptions and future flaring but also paves the way for a lower carbon footprint per barrel produced.

Gas to Power. Monetizing associated gas through gas-to-power projects or sales reduces flaring and generates revenue. In Field 2, partial gas compression and export led to a measurable reduction in flaring emissions and improved production efficiency.

Venting Reduction

Low-Emission Valves and Equipment. Replacing conventional valves and seals with low-emission alternatives reduces fugitive methane emissions. These upgrades are particularly effective in high-pressure systems and aging infrastructure.

Nitrogen for gas blanketing. In the oil industry, reinjection of water must adhere to stringent quality standards, particularly the exclusion of dissolved oxygen, which can compromise reservoir integrity and stimulate unwanted microbial growth. In Field 2, production gas was originally used for gas blanketing in the water treatment plant to maintain anaerobic conditions and prevent oxygen infiltration. While this approach effectively preserved water quality, it resulted in the continuous release of production gas into the atmosphere. To overcome this environmental drawback and enhance operational efficiency, the facility transitioned to nitrogen blanketing. This improvement not only elevated the quality of treated water destined for reinjection but also achieved a zero-emission outcome by fully eliminating atmospheric gas releases.

Methane Monitoring Technologies. Drone-based methane surveys provide critical data for identifying and quantifying venting sources. These technologies support targeted mitigation and validate emissions inventories, enhancing transparency and accuracy ([International Energy Agency, 2025](#))

Decarbonization Plan

A targeted decarbonization plan should prioritize actions that address the primary sources of emissions, while also delivering operational benefits. Building on the results summarized in [Table 11](#), recommended actions for each field are as follows:

Field 1:

- **Flare Reduction:** At the central processing facility, 1.2 mmscf of gas is currently flared. This gas could be redirected for power generation within the field's electrical grid, reducing reliance on oil or diesel as fuel elsewhere. Such a project would yield significant cost savings on diesel, enhance revenue by freeing more oil for sale, and reduce total field emissions by an estimated 25%. Given the relatively modest volumes, additional economic analysis is warranted.
- **Expanded Gas Utilization:** The gas flared at Well Pad 1 could also be routed to the central processing facility, increasing total gas utilization and further improving project economics.
- **High-Efficiency Pumps:** Upgrading pumps for the highest-producing wells (where 30% of wells account for 52% of production) could achieve a 5% reduction in overall field energy use.
- **Low-Venting Equipment and Nitrogen Blanketing:** Implementing low-emission equipment and using nitrogen for gas blanketing may yield an additional 4% emissions reduction.

- Enhanced Operational Practices: Implementing best practices during production and well interventions will help minimize fugitive emissions.

Field 2:

- Flare Reduction and Gas Utilization: There is notable potential to reduce flaring by expanding compression capacity and delivering more gas to the point of sale, with the potential to cut emissions by 18%. Gas-to-power initiatives could offer further reductions.
- High-Efficiency Pumps: Installing efficient pumps in the highest-producing wells may reduce energy consumption by up to 8%.
- Water Management: Adopting water shut-off and conformance technologies will help control subsurface water production, reducing both the energy required and the costs associated with chemical treatment and pump maintenance.
- Enhanced Operational Practices: As in Field 1, improving procedures during both routine operations and well interventions will further limit fugitive emissions.

Conclusions

- The structured methodology not only facilitates alignment with industry best practices and regulatory frameworks, but also serves as a catalyst for unlocking new opportunities in operational efficiency and sustainability. By adopting this robust tool for technical evaluation and continuous improvement, organizations position themselves to proactively discover innovative pathways for emissions reduction and value creation across diverse upstream oil and gas fields.
- The framework showcased in this manuscript reflects both technical rigor and practical adaptability, enabling seamless integration into existing workflows and rapid deployment. Its step by step approach encourages cross-functional collaboration and empowers operational teams to take ownership of emissions management as a strategic opportunity for growth, rather than a mere compliance requirement.
- Fully consistent with leading industry reporting protocols—including the Oil and Gas Climate Initiative (OGCI) guidelines—the methodology supports standardized quantification and transparent disclosure of greenhouse gas (GHG) emissions. This not only fulfills regulatory expectations, but also strengthens stakeholder confidence and enhances organizational reputation.
- Direct engagement of field-level personnel in implementing this methodology fosters a culture of innovation and continuous improvement. Through precise identification and prioritization of emission sources such as flaring, venting, and energy consumption, operational teams are equipped to drive impactful decarbonization initiatives and uncover additional avenues for optimization.
- The analytical framework further enables the development of customized decarbonization plans that are sensitive to field-specific operational realities and constraints. By quantifying the tangible benefits of targeted interventions—including flare reduction, high-efficiency pumping, water management, and enhanced operational practices—stakeholders can pursue actionable and economically attractive strategies for significant GHG reductions.
- Beyond emissions abatement, this methodology creates a platform for evaluating associated benefits like cost savings, increased sales revenue, operational reliability, and reduced compliance risk. Decision-makers can leverage these insights to realize the full value proposition of decarbonization, transforming emissions management into a driver of sustainable operational excellence and long-term competitiveness.
- In embracing this analysis, organizations are well positioned to turn challenges into opportunities and to lead the way in responsible, forward-thinking energy production.

References

- Abdul Muqtadir Khan, C. A. (2024). A Comprehensive Life Cycle Assessment of Hydraulic Fracturing. SPE International Health, Safety, Environment and Sustainability Conference and Exhibition. Abu Dhabi, UAE: SPE. doi: <https://doi.org/10.2118/220480-MS>
- Adam R. Brandt, M. S.-H. (2022). *Oil Production Greenhouse Gas Emissions Estimator*.
- Alexey Ruzhnikov, S. D., Aleksandr Nifantov, S. P., & Mohammed. (2023). The Accurate Emission Forecast at the Well Planning Stage. *ADIPEC*. Abu Dhabi, UAE. doi: <https://doi.org/10.2118/216516-MS>
- API. (2021, November). <https://www.api.org/news-policy-and-issues/news/2021/11/10/api-updates-ghg-compendium>. Retrieved from Compendium of GHG Emissions Methodologies for the Natural Gas and Oil Industry: <https://www.api.org/~media/Files/Policy/ESG/GHG/2021-API-GHG-Compendium-110921.pdf>
- Arthur, B. (2008). The Relation between LCA and GHG Accounting. *34th Swiss LCA Discussion Forum 'LCA vs. GHG footprint'*. Zurich: E2 Management Consulting AG.
- Augusto Correnti, D. A. (2022). Early production Facility Energy Efficiency Analysis. *11° Congreso de Exploración y Desarrollo de Hidrocarburos Desarrollo y Sustentabilidad*. Mendoza, Argentina: IAPG. Retrieved from <https://www.iapg.org.ar/conexplo/PENDRIVE/pdf/sesiones-generales/desarrollo-sustentabilidad/desarrollo-y-sustentabilidad11.PDF>
- Carpenter, C. (2024, February). Rig Automation Empowers. *JPT*. doi: <https://doi.org/10.2118/0224-0052-JPT>
- César Alejandro Briones Luna, C. V. (2022). Well Production Optimization through Annulus Gas Management using Gas Handling System in Auca Field, Ecuador. *SPE Western Regional Meeting*. Bakersfield, California: SPE. doi: <https://doi.org/10.2118/209339-MS>
- Chandharabhanich, I., Mekkerdchoo, B., Pensook, T., Amranand, S., Dejdamrongprecha, W., Jiratawaree, S., ... Sairuk, C. (2024). Enhancing Fuel Efficiency in Onshore Drilling Rigs through Generator Power Management: A Sustainable Approach to Reduce Diesel Consumption. Kuala Lumpur, Malaysia: SPE. doi: <https://doi.org/10.4043/34724-MS>
- Climate & Clean Air Coalition OGMP. (2015). *Technical Guidance Document Number 9: Casing Head Venting*. Retrieved from www.ccacoalition.org: https://www.ccacoalition.org/sites/default/files/resources/2017_OGMP-TGD9-Casinghead-venting_CCAC.pdf
- Dmitry Pleshkov, C. A. (2022). Disruptive Dumpflooding Completion, Case Study Ecuador. Offshore Technology Conference Asia, Virtual and Kuala Lumpur, Malaysia, Kuala Lumpur, Malaysia: SPE. doi: <https://doi.org/10.4043/31533-MS>
- Environmental Ministry. (n.d.). reporte. Quito. Retrieved from <https://www.bing.com/search?q=factor%20de%20emisiones%20sistema%20interconectado%20nacional%20ecuador%202024&qsn&form=QBRE&sp=-1&lq=0&pq=factor%20de%20emisiones%20sistema%20interconectado%20nacional%20ecuador%202024&sc=10-64&sk=&cvid=18D2E848C7D84763A433>
- Etcheverry, K., Peña, K., Quevedo, H., Rojas, E., Correa, R., Castellanos, J., ... Goodkey, B. (2023). Digital Innovation of Well Construction Process in Ecuador Through Rig Automation. *ADIPEC*. Abu Dhabi, UAE, SPE. doi: <https://doi.org/10.2118/216249-MS>
- Green House Gas Protocol. (n.d.). *A Corporate Accounting and Reporting Standard*. Retrieved from Green House Gas Protocol: <https://ghgprotocol.org/>
- International Energy Agency. (2023). *Emissions from Oil and Gas Operations in Net Zero Transitions*. International Energy Agency. Retrieved from <https://iea.blob.core.windows.net/assets/f065ae5e-94ed-4fcb-8f17-8ceffde8bdd2/TheOilandGasIndustryinNetZeroTransitions.pdf>
- International Energy Agency. (2025). *Global Methane Tracker 2025*. Paris: IEA. Retrieved from <https://www.iea.org/reports/global-methane-tracker-2025>
- IPCC. (2023, December 30). <https://catalog.data.gov/dataset/ipcc-ar4-ar5-and-ar6-20-100-and-500-year-gwps>. doi: <https://doi.org/10.23719/1529821>
- ISO-14040. (2006). ISO-14040 Environmental management — Life cycle Assessment - Principles and Frameworks. International Standard Organization.
- Luis Tilleria, G. A. (2025). Energy Balance in the Oil Sector: Innovation and Sustainability. *ADIPEC*. Abu Dhabi: SPE.
- M. N. Saputra, P. S., & B. A. Rahmanto and H. Anggaraini. (2025). Enhancing Reliability and Emissions Reduction: The Impact of Compact Wide Range Electrical Submersible Pumps. International Petroleum Technology Conference h. n Kuala Lumpur, Malaysia, 1. doi: <https://doi.org/10.2523/IPTC-24925-EA>
- Nicholson, S. a. (2021, January 21). *Life Cycle Emissions Factors for Electricity Generation Technologies*. doi: [10.7799/1819907](https://doi.org/10.7799/1819907)
- Nifantov, A., & Ruzhnikov, A. (2023). Well Design Re-Engineering: First Level of Emission Footprint Optimization. *Middle East Oil, Gas and Geosciences Show*, Manama, Bahrain, SPE. doi: <https://doi.org/10.2118/213365-MS>

- Nifantov, A., Ruzhnikov, A., Corona, M., Alharthi, R., Hussein, A. N., & Chandran, K. J. (2024). Application of Drilling Technologies to Manage Emission Footprint During Well Construction. SPE International Health, Safety, Environment and Sustainability Conference and Exhibition. Abu Dhabi, UAE,: SPE. doi: <https://doi.org/10.2118/220457-MS>
- OGCI. (2024, March). OGCI Reporting Framework. Retrieved from *OGCI Performance Data hub*: https://www.ogci.com/wp-content/uploads/2024/11/OGCI_Reporting_Framework_2024_Final.pdf
- Oil and Gas Decarbonization Chapter. (2024). The Oil and Gas Decarbonization Chapter: 2024 Baseline. *Oil and Gas Decarbonization Chapter*. OGDG. Retrieved from <https://www.ogdc.org/wp-content/uploads/2024/11/The-Oil-Gas-Decarbonization-Charter-Report-2024-1.pdf>
- Paola Rueda, H. Q. (2023). Smart Production Operations in a Remote Field in Latin America. *ADIPEC*. Abu Dhabi, UAE: SPE. doi: <https://doi.org/10.2118/216827-MS>
- Piñeiros, S., Mendoza, M. A., Cevallos, G., Molina, E., Azancot, A., & Bermeo, P. (2020). An Integrated New Exploitation Strategy in Bloque 61, from Modeling to Full-Field Implementation. SPE Latin American and Caribbean Petroleum Engineering Conference. SPE. doi: <https://doi.org/10.2118/198964-MS>
- Ruiz, Y. L., Carrera, J. M., Tagarot, G. N., Gey, G. M., Davalos, D., Segovia, R. D., ... Pastrana, W. A. (2022). Transforming Legacy Production Operations into Smart Asset Operations in Ecuador. *ADIPEC*. Abu Dhabi, UAE: SPE. doi: <https://doi.org/10.2118/211844-MS>
- Santiago Davalos, H. Q. (2024). Innovative and Integrated Digital Solution for Water-Handling Management System Using Machine Learning for Production Enhancement in a Brownfield. SPE Water Lifecycle Management Conference and Exhibition,. doi: <https://doi.org/10.2118/219005-MS>
- Sterling, A. M., Ruiz, Y. L., Azancot, A., Charry, L., Correal, A., Fuentes, L., ... Vasconez, N. D. (2020). Economical and Technical Solution for Waterflooding Projects in a Mature Field: Modified Dump Flooding Well Completion Strategy. SPE Latin American and Caribbean Petroleum Engineering Conference. SPE. doi: <https://doi.org/10.2118/198979-MS>
- World Bank Group. (2016). *Global gas flaring reduction partnership - gas flaring definitions (English)*. Washington, D.C: World Bank Group. Retrieved from Global gas flaring reduction partnership - gas flaring definitions (English).: <http://documents.worldbank.org/curated/en/755071467695306362>
- World Bank Group. (2025). *2025 Global Gas Flaring Tracker Report*. 2025 Global Gas Flaring Tracker Report. Retrieved from <https://thedocs.worldbank.org/en/doc/bd2432bbb0e514986f382f61b14b2608-0400072025/original/Global-Gas-Flaring-Tracker-Report-July-2025.pdf>